REFLOW.

A technical case study · Jasper Middendorp, Co-founder & CEO

FROM WASTE TO FILAMENT

Turning plastic waste into 3D-printing filament.

Reflow was an Amsterdam social enterprise that converted recycled plastic — gathered by waste pickers — into high-quality 3D-printing filament. This is the technical story of the process, the materials, and the impact.

2015	95%+	2
FOUNDED	RECYCLED CONTENT	FILAMENT FORMATS

The venture

A circular-economy startup building a new supply chain for desktop 3D printing — one that pays waste collectors and keeps plastic out of landfill.

Founded in 2015 and on the market from 2019 out of Amsterdam, Reflow made desktop 3D-printing filament from recycled plastic. The model paired a genuinely technical challenge — producing dimensionally stable, printable filament from heterogeneous waste — with a social one: sourcing that plastic from waste pickers in places like India, turning collection into fair, traceable income rather than commodity extraction.

The problem

Desktop 3D printing ran almost entirely on virgin plastic while the world drowned in unrecycled PET. Closing that loop meant solving material consistency — the hard part of recycling waste into a precision consumable.

The model

A social-enterprise supply chain: plastic collected by waste pickers → processed into filament → sold to makers and industry. Impact embedded in the product, not bolted on.

Circular economy · Social enterprise · Made in NL

The process — waste to filament

Six stages turn post-consumer plastic into a precision 3D-printing consumable. The difficulty lives in stages 4–6: holding diameter and moisture tolerances on feedstock that is, by definition, inconsistent.

01
Collect
Post-consumer plastic (PET bottles, PLA scrap) sourced from waste pickers with traceable, fair payment.

02
Sort
Separate by polymer type and color; reject contaminants. Polymer purity drives everything downstream.

03
Shred
Flake the sorted plastic into uniform chips — consistent flake size keeps extruder feed steady.

04
Extrude
Melt-extrude flake through a precision die to 1.75 mm / 2.85 mm filament; cooled and pulled at controlled rate.

05
QA
Laser diameter measurement and feedback control to hold tolerance; batch tracking for printability.

06
Spool
Wind onto 1 kg spools, vacuum-seal against moisture pickup, label with material and batch.

Recycled PET is hygroscopic and crystalline — it prints hotter than PETG, stringing and moisture absorption are constant adversaries. Diameter consistency (target ±0.05 mm on 1.75 mm) is the single largest quality lever, since extrusion flow scales with the square of filament diameter.

Materials & specifications

Material	Source	Diameter	Tolerance	Recycled	Notes
rPET	Recycled PET bottles	1.75 / 2.85 mm	~±0.05 mm	95%+	Stiff, strong; hotter print than PETG; keep dry.
Recycled PLA (Matte)	Mechanically recycled PLA	1.75 / 2.85 mm	~±0.05 mm	95%+	Biodegradable/renewable feedstock; matte finish.
rPET (industrial)	Industrial plastic waste	1.75 / 2.85 mm	~±0.05 mm	up to 100%	Engineered for 3D-printed tooling, incl. aviation jigs.

Applications & impact

Where the filament went, and what the model achieved beyond the kilogram.

Applications

General desktop printing for makers and education, and higher-spec 3D-printed **tooling** — jigs, fixtures and prototypes, including for aviation — where recycled feedstock met the performance bar for functional parts.

Makers · Education · Industrial tooling · Aviation jigs

Impact

Plastic diverted from landfill or incineration, and — the core of the model — fair, traceable income for waste pickers who collected the feedstock. An award-winning social enterprise approach to circular manufacturing.

Diverted waste · Waste-picker livelihoods · Award-winning

Founding team

Lyndsey Lewis · Jasper Middendorp · Rahul Mehendiratta · Ronan Hayes — Amsterdam, NL.

Publications, talks & press

Selected coverage, talks, and profiles of Reflow and Jasper Middendorp's work.

TALK · INSIDE 3D PRINTING

Reflow presentation at Inside 3D Printing

pt.slideshare.net/.../reflow-presentation-at-inside3dprinting-conference

ARTICLE · MEDIUM

Start with what's already there — Reflow's approach to impact

medium.com/.../reflow-s-approach-to-making-an-impact

FEATURE · UC BERKELEY FUNG INSTITUTE

Reflow Filament: empowering communities through sustainability

funginstitute.berkeley.edu/.../reflow-filament-empowering-communities

INTERVIEW · EMERCE (NL)

Start-up Reflow: van afval naar 3D-printmateriaal

emerce.nl/.../start-up-reflow-van-afval-naar-3d-printmateriaal

SPOTLIGHT · WEWORK

Help Reflow turn scrap plastic into a hot commodity

wework.com/.../amsterdams-reflow-turns-scrap-plastic-hot-commodity

PRESS · RTL (NL)

De startup Reflow maakt banen met plastic afval

rtl.nl/.../de-startup-reflow-maakt-banen-met-plastic-afval

PROFILE · CRUNCHBASE

Jasper Middendorp — Founder & CEO, Reflow

crunchbase.com/person/jasper-middendorp

PROFILE · F6S

Jasper Middendorp — CEO & Co-founder, Reflow

f6s.com/member/jaspermiddendorp1

Related work: at AMS Institute, Jasper also led **Junai** — paste-extrusion 3D printing from organic waste — and serves on the board of **Junai.earth** (home-compostable 3D printing).

REFLOW.

Building things that matter, across software and the physical world.

Reflow was a chapter in a hands-on career building products end-to-end — from recycled-filament manufacturing to the AI-native software studio that followed. Same through-line: pick a hard, useful problem and ship it.

linkedin.com/in/jaspermiddendorp

noboxdev.com

github.com/No-Box-Dev